

2. Class Diagram Problem - Library Management System

a) Identify classes with attributes and methods

Classes involved:

- Book
- Member
- Librarian
- IssueRecord

Each class contains relevant attributes and operations as shown in the class diagram.

b) Define relationships (association, aggregation, inheritance)

- Association: Member is associated with IssueRecord.
- Association: Book is associated with IssueRecord.
- Association: Librarian is associated with IssueRecord.
- Aggregation: Library Management System consists of Book, Member, Librarian and IssueRecord.
- Inheritance: Member and Librarian can be considered specialized users of the Library Management System.-

c) Specify multiplicity between classes

- One Member can have many IssueRecords (1 to 0..*).
- One Book can appear in many IssueRecords over time (1 to 0..*).
- One Librarian can manage many IssueRecords (1 to 0..*).

d) Class Diagram

